

Molecular transition 基準 (primary line lists; 代表 transitions)

H2 (ExoMol RACPPK; top 10 by A)

- 1) $v=0, J=28 \rightarrow v=0, J=26$
 $\tilde{\nu}=2903.822736 \text{ cm}^{-1}$ $\lambda=3.444 \text{ }\mu\text{m}$ $A=5.506\text{e-}06 \text{ s}^{-1}$
- 2) $v=0, J=27 \rightarrow v=0, J=25$
 $\tilde{\nu}=2931.587453 \text{ cm}^{-1}$ $\lambda=3.411 \text{ }\mu\text{m}$ $A=5.503\text{e-}06 \text{ s}^{-1}$
- 3) $v=0, J=29 \rightarrow v=0, J=27$
 $\tilde{\nu}=2869.043657 \text{ cm}^{-1}$ $\lambda=3.485 \text{ }\mu\text{m}$ $A=5.430\text{e-}06 \text{ s}^{-1}$
- 4) $v=0, J=26 \rightarrow v=0, J=24$
 $\tilde{\nu}=2952.202695 \text{ cm}^{-1}$ $\lambda=3.387 \text{ }\mu\text{m}$ $A=5.424\text{e-}06 \text{ s}^{-1}$
- 5) $v=0, J=30 \rightarrow v=0, J=28$
 $\tilde{\nu}=2827.268848 \text{ cm}^{-1}$ $\lambda=3.537 \text{ }\mu\text{m}$ $A=5.275\text{e-}06 \text{ s}^{-1}$
- 6) $v=0, J=25 \rightarrow v=0, J=23$
 $\tilde{\nu}=2965.440121 \text{ cm}^{-1}$ $\lambda=3.372 \text{ }\mu\text{m}$ $A=5.274\text{e-}06 \text{ s}^{-1}$
- 7) $v=0, J=24 \rightarrow v=0, J=22$
 $\tilde{\nu}=2970.994204 \text{ cm}^{-1}$ $\lambda=3.366 \text{ }\mu\text{m}$ $A=5.056\text{e-}06 \text{ s}^{-1}$

Selection: top-N by Einstein A.
Target fix only (not P-model derivation).

HD (ExoMol ADJSAAM; top 10 by A)

- 1) $v=9, J=23, \text{qb} \rightarrow v=7, J=22, \text{b}$
 $\tilde{\nu}=3246.097634 \text{ cm}^{-1}$ $\lambda=3.081 \text{ }\mu\text{m}$ $A=1.188\text{e-}03 \text{ s}^{-1}$
- 2) $v=3, J=33, \text{b} \rightarrow v=2, J=32, \text{b}$
 $\tilde{\nu}=2657.952656 \text{ cm}^{-1}$ $\lambda=3.762 \text{ }\mu\text{m}$ $A=1.133\text{e-}03 \text{ s}^{-1}$
- 3) $v=9, J=23, \text{qb} \rightarrow v=5, J=24, \text{b}$
 $\tilde{\nu}=4996.782991 \text{ cm}^{-1}$ $\lambda=2.001 \text{ }\mu\text{m}$ $A=2.805\text{e-}04 \text{ s}^{-1}$
- 4) $v=9, J=23, \text{qb} \rightarrow v=6, J=22, \text{b}$
 $\tilde{\nu}=4953.068383 \text{ cm}^{-1}$ $\lambda=2.019 \text{ }\mu\text{m}$ $A=2.728\text{e-}04 \text{ s}^{-1}$
- 5) $v=0, J=25, \text{b} \rightarrow v=0, J=24, \text{b}$
 $\tilde{\nu}=1287.580183 \text{ cm}^{-1}$ $\lambda=7.767 \text{ }\mu\text{m}$ $A=1.864\text{e-}04 \text{ s}^{-1}$
- 6) $v=5, J=17, \text{b} \rightarrow v=4, J=16, \text{b}$
 $\tilde{\nu}=3346.094087 \text{ cm}^{-1}$ $\lambda=2.989 \text{ }\mu\text{m}$ $A=1.859\text{e-}04 \text{ s}^{-1}$
- 7) $v=5, J=18, \text{b} \rightarrow v=4, J=17, \text{b}$
 $\tilde{\nu}=3311.534921 \text{ cm}^{-1}$ $\lambda=3.020 \text{ }\mu\text{m}$ $A=1.858\text{e-}04 \text{ s}^{-1}$

Selection: top-N by Einstein A.
Target fix only (not P-model derivation).

D2 (MOLAT ARLSJ1999 (FUV; B/C/B'/D→X); top 10 by A)

- 1) $\text{C}-(v=11, J=1) \rightarrow \text{X}(v=17, J=1)$
 $\tilde{\nu}=79256.690000 \text{ cm}^{-1}$ $\lambda=0.126 \text{ }\mu\text{m}$ $A=5.274\text{e+}08 \text{ s}^{-1}$
- 2) $\text{C}-(v=11, J=2) \rightarrow \text{X}(v=17, J=2)$
 $\tilde{\nu}=79251.560000 \text{ cm}^{-1}$ $\lambda=0.126 \text{ }\mu\text{m}$ $A=5.274\text{e+}08 \text{ s}^{-1}$
- 3) $\text{C}-(v=11, J=3) \rightarrow \text{X}(v=17, J=3)$
 $\tilde{\nu}=79244.160000 \text{ cm}^{-1}$ $\lambda=0.126 \text{ }\mu\text{m}$ $A=5.268\text{e+}08 \text{ s}^{-1}$
- 4) $\text{C}-(v=11, J=4) \rightarrow \text{X}(v=17, J=4)$
 $\tilde{\nu}=79235.040000 \text{ cm}^{-1}$ $\lambda=0.126 \text{ }\mu\text{m}$ $A=5.248\text{e+}08 \text{ s}^{-1}$
- 5) $\text{C}-(v=11, J=5) \rightarrow \text{X}(v=17, J=5)$
 $\tilde{\nu}=79224.640000 \text{ cm}^{-1}$ $\lambda=0.126 \text{ }\mu\text{m}$ $A=5.205\text{e+}08 \text{ s}^{-1}$
- 6) $\text{C}-(v=13, J=10) \rightarrow \text{X}(v=18, J=10)$
 $\tilde{\nu}=80239.490000 \text{ cm}^{-1}$ $\lambda=0.125 \text{ }\mu\text{m}$ $A=5.171\text{e+}08 \text{ s}^{-1}$
- 7) $\text{C}-(v=11, J=6) \rightarrow \text{X}(v=17, J=6)$
 $\tilde{\nu}=79213.910000 \text{ cm}^{-1}$ $\lambda=0.126 \text{ }\mu\text{m}$ $A=5.123\text{e+}08 \text{ s}^{-1}$

Selection: top-N by Einstein A.
Target fix only (not P-model derivation).